

**MULTILINGUAL SCAM SMS AND CALL DETECTION
WITH AWARENESS SUPPORT SYSTEM USING
NLP AND MACHINE LEARNING**

A PROJECT REPORT

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in partial fulfillment for the award of the

degree of

BACHELOR OF TECHNOLOGY

IN

COMPUTER SCIENCE AND BUISNESS SYSTEMS



AKSHAYA COLLEGE OF ENGINEERING AND TECHNOLOGY

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**Bhagavathipalayam, Kinathukadavu, Coimbatore,
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MAY 2026

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ACKNOWLEDGEMENT

With genuine humility, we are obediently thankful to God Almighty without Him, this work would have never become a reality.

We express our heartfelt thanks to our **Joint Director, Dr. N. SUGUNA, M.E., Ph.D.**, for her continuous encouragement, valuable support, and motivation throughout the course of this project.

We are deeply grateful to our **CEO, Dr. S. KAPILAN, M.E., Ph.D.**, whose visionary leadership and constant support have been a driving force in our academic and project endeavors.

We wish to express our sincere gratitude to our respected **Principal, Dr. R. RAVINDRAN, M. Tech., Ph.D.**, for providing us with the necessary resources and opportunity to successfully complete our project work.

We are thankful to our **Vice Principal, Dr. K. SIVASANKARI, M.E., Ph.D.**, for her insightful feedback and valuable guidance that contributed significantly to the progress of our project.

We extend our deep sense of gratitude to our beloved **Dr. M. MALINI, M. Tech., Ph.D.**, Associate Professor and Head, Department of Computer Science and Business Systems who has been a constant source of motivation and instrumental in broadening our knowledge and understanding.

Our sincere appreciation goes Project Guide, **Mr.S.BALAJI, M.E.**, Assistant Professor, Department of Computer Science and Business Systems for his dedicated mentorship, timely feedback, and continuous support throughout the development of our project.

We also thank the project review members and all the faculty members of the Department of Computer Science and Business Systems for their valuable contributions to our learning journey.

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ABSTRACT

Due to the expanding use of digital communication channels, there is an increased emphasis on detecting fraudulent SMS messages and scam phone calls for purposes of user safety and data integrity. Historically, the method of detection was accomplished through rule-based filtering, lacked context understanding, and was heavily reliant on user judgement, resulting in unreliable and inconsistent outcomes. Furthermore, both traditional and automated detection methods have difficulty with multiple languages and are not able to adapt to changing methods of scamming, resulting in many systems being ineffective in a practical application. Automated detection systems utilizing both Artificial Intelligence and Natural Language Processing offer an effective and scalable alternative for detecting fraudulent communication through objective and data-based identification of fraud. The following describes the design and implementation of a Multilingual Scam SMS and Call Detection System, an intelligent web-based platform developed using a Flask framework, which automatically analyses and classifies messaging through methods of machine learning. The system will process both SMS and voice calls by transforming the information into a structured text format, and by utilizing machine learning classification models to provide a prediction for fraud. The fraudulent SMS and phone call detection system are capable of detecting fraud in both text and voice communications, by utilizing TF-IDF for feature extraction, and Naive Bayes classifier for prediction purposes, as well as text to speech functionality and multilingual detection support, improving accuracy, reliability, and user awareness.

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LIST OF ABBREVIATIONS

ABBREVIATION	FULL FORM
AI	Artificial Intelligence
ML	Machine Learning
NLP	Natural Language Processing
TF-IDF	Term Frequency–Inverse Document Frequency
NB	Naive Bayes
STT	Speech-To-Text
SMS	Short Message Service
UI	User Interface
DB	Database
API	Application Programming Interface
HTML	Hyper Text Markup Language
CSS	Cascading Style Sheets
JS	JavaScript
GUI	Graphical User Interface
SVM	Support Vector Machine
ANN	Artificial Neural Network
CSV	Comma Separated Values
URL	Uniform Resource Locator
JSON	JavaScript Object Notation
IDE	Integrated Development Environment
VS	Visual Studio

CHAPTER 1

INTRODUCTION

1.1 OVERVIEW

The rapid advances of digital communications have made SMS and voice telephone communications a vital means of interaction in our daily lives. However, the increased use of these communication means has led to a rise in construction fraud with an increase in issues such as scam messages and fraud through telephone calls. Scams take the form of acquisition of your personal and financial information by fraudulent means such as phishing links, fictitious offers, impersonation and financial fraud. As such, the user will find it very challenging to determine if a specific message or call is fraud.

Traditional systems for detecting these types of scams rely on rule-based filtering and manual judgment. These systems have limitations in accuracy, scalability and generally have limitations in detecting patterns for new types of scams especially when using multiple languages.

The advances of Artificial Intelligence and Natural Language Processing have provided us with the tools to develop automated systems that can detect scams. This project is developing a Multilingual Scam SMS & Call Detection System which will utilize the TF-IDF statistical measure to extract features of SMS messages and classify these features using a Naive Bayes classifier. The system will also employ speech to text technology to analyze calls and employ an awareness module to provide users with knowledge of potential scam activity. All these elements will help to improve the accuracy of the system, the ability of the system to detect scams in real-time, and enhance user security from scams using digital technologies.

1.2 PROBLEM STATEMENT

Detection systems for scams have been limited by their ability to identify fraudulent SMS texts and phone calls with high levels of accuracy, consistency and efficiency. Most traditional detection systems rely heavily on a rule-based filtering system, as well as on manual judgement from users, leading to subjective and inconsistent reliability. Users of conventional detection systems may interpret the same message differently, negatively impacting the consistency of identifying messages that are scams. Because of the lack of standardized evaluation techniques, it is also difficult for traditional detection systems to accurately classify the same messages when using different formats and patterns.

Another large problem for conventional detection systems is that the systems do not detect frauds in two languages, or scams committed in a voice format. The majority of detection systems is built primarily to detect fraud in the English language, and will typically struggle to detect scams that were sent in a local or mixed-language format. In addition, very few detection systems provide real-time detection and/or structured feedback to users about risks, as well as about patterns associated with scams.

As scams continue to grow and become more sophisticated, so do scammers' methods of operation; this makes it increasingly difficult to detect scams. Furthermore, traditional detection systems lack machine learning technology that would allow them to adapt to new patterns of operation. Therefore, there is a substantial need for a system that uses automation, scalability, intelligence and technology to effectively identify scams, and improve user awareness and protection across all forms of communication.

1.3 USING ML ENSEMBLE METHODS

By using both natural languages processing and machine-learning, the proposed system is able to recognize bogus text messages and telephone calls. SMS messages are processed first using a series of pre-processing steps e.g., tokenization, stop words and normalization followed by text analysis.

After this initial step, the SMS messages are then extracted from the features of the message using Term Frequency-Inverse Document Frequency to identify the key words/phrases in the message that are indicative of the nature of the scam.

Messages will be classified as either spam or legitimate using a Naïve Bayes classifier which is built from training examples. When it comes to phone scams, the proposed system employs speech to text technology in order to translate spoken words into written words/to text so that they may be analyzed as part of the text analysis of the SMS messages, both sources of data SMS and telephone will be treated in the same manner.

The proposed system will provide support for multi-lingual data in order to identify scams in various languages as well as particular regional contexts therefore adding capabilities to the detection system by providing the system with context surrounding the entire message and to provide an understanding of the scams being committed.

The proposed system affords much less reliance on pre-scripted authenticating rules than traditional rule-based detection systems. Additionally, due to the use of machine learning ML and NLP, it will provide an accelerated ability to recognize.

1.4 OBJECTIVE OF THE PROJECT

The objectives of the project are to design and develop an intelligent system capable of detecting fraudulent SMS messages and phone call scams using real-time data and machine learning techniques.

- To develop a web-based platform that analyses SMS messages and call transcripts for scam detection in real-time.
- To extract and process textual features using TF-IDF and apply machine learning models such as Naive Bayes for classification.
- To classify messages and calls into Safe, Suspicious, and Scam categories to improve detection accuracy.
- To integrate speech-to-text technology for detecting voice-based scams from call audio.
- To support multilingual communication and provide user awareness for preventing fraud.

Thus, the objective of this project is to develop a reliable, scalable, and intelligent system for real-time scam detection and prevention.

1.5 SCOPE OF THE PROJECT

Multilingual Scam SMS and Call Detection System is an intelligent automated platform that detects fraudulent communications through machine learning and natural language processing techniques. The platform performs this function by converting both SMS and voice call communication into structured text for easier processing. It gives people information in almost real time regarding the classification of the received SMS message or voice call as being safe, suspicious

or a scam allowing for better informed decision making and higher quality digital security.

The project uses several types of ML algorithms such as TF-IDF for feature extraction and the Naïve Bayes algorithm for accurate prediction. Additionally, voice-to-text technology is used to convert call conversations to text to allow for the unified processing of both text and voice data. In order to help detect scams across a broad range of users, the system has the capacity to handle communications in English, Tamil and Hindi. The platform also contains an awareness module to educate users about how to identify common scam patterns and how to protect themselves from becoming victims of a scam.

The system is built on the Flask platform and has a user-friendly interface for interacting with the system. The system's main goals are to improve the accuracy of scam detection, reduce the amount of manual effort required by the users to determine if a scam message is present, and implement solutions that allow the scalable deployment of the scam detection capabilities of the system in a real-world communications systems.

CHAPTER 2

LITERATURE REVIEW

2.1 SMISHING DETECTION FROM A MESSAGING PLATFORM VIEW

A machine learning-based smishing detection system was proposed by Schwarz S. F. et al. [1], where SMS messages are analysed using textual patterns and keyword-based classification methods. The system identifies fraudulent messages by examining suspicious URLs and scam-related keywords, improving automated filtering and detection accuracy. However, the system is limited to SMS-only data and supports only English, reducing its effectiveness in multilingual environments. It cannot detect voice-based scams and lacks adaptability to evolving scam techniques.

2.2 SMS SPAM DETECTION USING RELEVANCE VECTOR MACHINE

A spam detection system using the Relevance Vector Machine (RVM) algorithm was proposed by Pudasaini S. et al. [2], focusing on classifying SMS messages using a probabilistic approach. The system improves accuracy and reduces false positives by analyzing textual features. However, it is limited to SMS data, lacks multilingual support, and cannot handle voice-based scams. It also requires labeled datasets and has higher computational complexity, making it less suitable for real-time applications.

2.3 FRAUDALENT E-WALLET SMS DETECTION

A machine learning-based system for detecting fraudulent e-wallet SMS messages was proposed by Enkono F. S. and Suresh N. [3], focusing on financial fraud detection in digital payment systems. The system analyzes transaction-related messages and identifies suspicious patterns using classification algorithms. It improves security by detecting fraudulent activities and preventing unauthorized transactions, enhancing trust in digital communication systems. However, the system is domain-specific and limited to e-wallet SMS detection, reducing its general applicability. It does not support multilingual communication and requires domain-specific training data. Additionally, it lacks voice-based scam detection and real-time processing capabilities.

2.4 SMISHING DETECTION VIA ML

A comprehensive review of machine learning techniques for smishing detection was presented by Mahmood A. R. and Hameed S. M. [4], where various algorithms are analyzed to evaluate their effectiveness in detecting scam messages. The study compares traditional machine learning models and advanced deep learning approaches based on accuracy, efficiency, and performance. It highlights key research trends and discusses important challenges such as feature extraction, dataset limitations, and evolving scam patterns. This work provides useful insights for researchers and helps in understanding existing detection techniques. However, the study does not propose a practical system and lacks real-time implementation. It does not support multilingual or voice-based detection and mainly focuses on theoretical analysis, limiting its real-world applicability.

2.5 HYBRID DL APPROACHES

A hybrid deep learning-based smishing detection system was proposed by Mahmud T. et al. [5], where multiple deep learning models are combined to improve classification accuracy. The system integrates techniques such as CNN and RNN to analyze both contextual and sequential patterns in text messages. It effectively detects complex scam patterns and adapts to evolving phishing techniques. The model performs well on large datasets and enhances detection capability by learning deep features automatically. However, the system requires high computational resources and large training datasets, making it expensive to implement. It is complex and difficult to interpret, and not suitable for lightweight or real-time applications, especially in resource-constrained environments.

2.6 DL APPROACH FOR SMISHING DETECTION

A deep learning-based smishing detection system was introduced by Mehmood M. K. et al. [6], focusing on improving detection accuracy and reducing false positives. The model uses neural networks to analyze textual data and identify hidden patterns in scam messages. It learns from large datasets and adapts to new and evolving phishing techniques, improving classification reliability. The system supports automated detection and enhances performance compared to traditional methods. However, it requires high computational power and large datasets for training, increasing implementation complexity. The model lacks explainability and is difficult to interpret, making it less transparent. It is also not suitable for lightweight or real-time applications.

2.7 SPAM EMAIL CLASSIFICATION USING ET

An ensemble-based spam classification system was proposed by Adnan M. et al. [7], combining multiple machine learning algorithms to improve detection accuracy. The system integrates different models to reduce errors and enhance overall performance. It improves generalization and handles complex datasets effectively, providing more reliable classification results. The approach increases prediction accuracy and reduces misclassification compared to single models. However, the system increases computational complexity and requires more processing time. It is difficult to implement and requires careful tuning of multiple models. Additionally, it is not suitable for light weight or real-time applications, limiting its practical usability.

2.8 SMISHING DETECTOR USING NEURAL NETWORKS

A neural network-based smishing detection system was proposed by Mishra S. and Soni D. (2022), focusing on identifying scam messages using deep learning techniques. The system learns complex patterns from textual data and improves detection accuracy. It supports automated classification and adapts to dynamic scam patterns, making it effective for evolving phishing attacks. The model enhances detection efficiency and provides intelligent decision-making. However, it requires large datasets for training and involves high training time. The system is computationally expensive and may suffer from overfitting issues. Additionally, it lacks interpretability, making it difficult to understand prediction results.

2.9 SMISHING DETECTION USING ML ALGORITHMS

A machine learning-based smishing detection system was proposed by Shaikh A. et al. [9], utilizing multiple algorithms for detecting phishing messages. The system compares model performance and improves classification accuracy. It enhances communication security and supports automated detection. However, it is limited to SMS data and lacks multilingual and voice-based detection. It also depends heavily on datasets and lacks real-time processing capability.

2.10 SMISHING DEFENSE USING META-HEURISTIC OPTIMIZATION

A smishing detection system using meta-heuristic optimization techniques was proposed by Alshinwan M. et al. [10], focusing on improving feature selection and classification performance. The system enhances accuracy by identifying important features and optimizing model parameters. However, it has high computational complexity and is difficult to implement. It requires expert knowledge and is not suitable for lightweight applications, with limited real-world deployment.

CHAPTER 3

SYSTEM ANALYSIS

3.1 EXISTING SYSTEM

Most existing scam detection systems rely on traditional rule-based techniques such as keyword filtering and blacklist-based approaches. These systems classify messages using predefined patterns rather than understanding the actual context of the message. As a result, they struggle to detect modern and evolving scam techniques, leading to missed fraudulent messages and incorrect classification of legitimate ones. This reduces the overall accuracy and reliability of the system.

In many situations, users must rely on their own judgment to determine whether a message or call is fraudulent. Suspicious activities are usually reported manually to service providers or authorities, making the process reactive rather than proactive. This delay increases the chances of financial loss, as there are no real-time automated detection systems to prevent scams immediately. The lack of instant response mechanisms limits the effectiveness of current solutions.

Additionally, current systems lack support for multilingual communication and voice-based scam detection. Most tools are designed for English text and cannot process regional languages or mixed-language messages effectively. They also do not include speech-to-text capabilities, making it difficult to analyze fraudulent calls. Furthermore, the absence of advanced machine learning techniques and user awareness features makes these systems less adaptable and less effective in handling modern scam scenarios.

3.2 PROPOSED SYSTEM

The Multilingual Scam SMS and Call Detection System is a user-friendly intelligent protection tool designed to detect and block fraudulent SMS and calls in real-time using NLP and ML technologies. The system analyzes the content of each SMS and transcription of each call to recognize patterns that may indicate spam or fraud. Fraudulent communications are categorized into three categories, Safe, Suspicious, or Scam, based on their recognized patterns. Because this is user-based, this system provides an additional layer of protection in addition to filtering tools available by platforms.

This system utilizes a lightweight solution leveraging TF-IDF and Naïve Bayes feature extraction to efficiently identify phishing keywords, malicious links and suspicious communication patterns at a low computational cost, thus providing a faster detection with higher accuracy, while being resource-constrained and therefore fitting well within typical real-time processing requirements.

By integrating a speech recognition technology layer, the system provides the same level of detection for voice communications as it does for messages. The system supports multiple languages including but not limited to English, Tamil and Hindi providing a wider range of usefulness. The system also has a module designed to educate users with alerts, safety tips and prevention recommendations while providing a log of all messages and calls analyzed by the system. All in all, the Multilingual Scam SMS and Call Detection System is a fast, dependable, user-friendly way to detect scam.

CHAPTER 4

SYSTEM DESIGN

4.1 GENERAL

The hardware/software components of developing and implementing the Multilingual Scam SMS and Call Detection System are discussed in this chapter. The proposed system is a user-level application developed using Natural Language Processing, Machine Learning, and Speech-to-Text technologies to detect scam messages and fraudulent calls in real time. The system analyzes SMS content and calls transcripts to classify them into Safe, Suspicious, or Scam categories while also providing user awareness. The hardware components support efficient execution of the application, while the software components support data processing, machine learning model implementation, and system deployment.

4.2 HARDWARE REQUIREMENTS

The hardware requirements for the proposed system are designed to support efficient execution of machine learning models, natural language processing, and real-time scam detection. A multi-core processor such as Intel i5 or AMD Ryzen 5 (or higher) is recommended to handle backend processing, feature extraction, and model inference. A minimum of 8 GB RAM is required to ensure smooth performance while processing SMS data, converting speech to text, and running the application simultaneously.

Storage requirements include at least 256 GB SSD to provide faster data access and reduce latency during system execution. Since the system analyzes SMS and call data and may use cloud-based APIs for speech recognition, a stable internet connection is essential for real-time processing and accurate detection.

4.3 SOFTWARE REQUIREMENTS

The software components play a vital role in the development, execution, and deployment of the Scam Detection system. Each software tool and technology is selected based on its efficiency, simplicity, and compatibility with machine learning workflows.

4.3.1 Flask Framework

Python is used as a main programming language for the development of this system because of its, readability and high availability of support for machine learning and NLP libraries. Due to the of implementation of the data processing algorithm, feature extraction algorithm and classification algorithms, Python can be integrated with Speech to Text tools and back-end development frameworks.

4.3.2 ML Methods

The core classification component of the system is based on the Naive Bayes algorithm, a probabilistic machine learning model widely used for text classification tasks. It works on the principle of conditional probability and assumes independence between features, making it computationally efficient and easy to implement. In this system, the Naive Bayes classifier is trained on labeled datasets containing both legitimate and scam messages, enabling it to learn patterns associated with fraudulent content. The model classifies input SMS messages and call transcripts into Safe, Suspicious, or Scam categories. Due to its lightweight nature and fast prediction capability, it is well-suited for real-time detection systems with limited computational resources.

4.3.3 Feature Extraction

TF-IDF is a method of creating vectors of numbers from textual data, which is capable of being processed by machine learning models. The technique provides a weighted value for each term, based upon the frequency of the term in one specific document and how often the term occurs across the entire set of documents that it is working with. Terms that appear frequently in a single document but do not appear as frequently in other documents are assigned greater importance. This may help determine key words that are representative of fraudulent activity, while minimizing the weight given to common words. Through the use of TF-IDF, unstructured text can be transformed into useful, numerical data for further processing and classification; therefore, TF-IDF is a fundamental part of increasing the accuracy and efficiency of the classification process.

4.3.4 Natural Language Processing Libraries

Tokenization, stop-word removal, text normalization, and keyword extraction are examples of the types of operations performed by Natural Language Processing libraries such as NLTK and spaCy. These tools are critical for preparing clean and well-structured data for input into machine learning models which ultimately improves the overall performance of a system.

4.3.5 Speech-To-Text Technology

The system has a Speech-to-Text function which allows it to convert voice calls into text for the purpose of analyzing them. This helps to detect scam text messages as well as in real time for voice calls. The text is processed using natural language processing techniques before being sent to a classifier machine learning model. By utilizing Speech to Text, the system is able to determine if there are fraudulent patterns found in spoken communications. Thus, Speech-to-Text enhances the overall capabilities of the system by allowing it to provide a more extensive means of detecting scams compared to just relying on traditional means of detecting scams through text only.

4.3.6 Frontend Technologies

To develop a system interface that is both interactive and easy to use, HTML CSS and JavaScript will be used for its design and development. HTML provides the structure of the WebPages; CSS provides the styles and design of the layouts to provide an attractive view; JavaScript provides dynamic functionality such as handling forms input, updating things in real-time and doing other activities that users interact with. Users can enter SMS messages, upload audio files of calls, and see the results of the classification through these technologies in an efficient manner. With an overall design that is easy to use and accessible for a breadth of users, this will create an easy to use system for all users.

4.3.7 SQLite Database

The database management system used to store and manage all of the system's information is SQLite. This is a lightweight, file-based, relational database that does not require an external server to run. The types of data stored in the SQLite

database include user input, processed messages, call transcripts, classification results, and system logs. SQLite provides efficient storage of data, fast access to it, and secure handling of that data. This simplicity and reliability make it an ideal choice for a large range of applications from small to medium sized plus they still provide enough performance for real-time data processing needs.

4.3.8 Visual Studio Code

Visual Studio Code is used as the development environment for implementing the Multilingual Scam SMS and Call Detection System. It is used to write and manage Python code for machine learning, NLP processing, and speech-to-text integration. Features like syntax highlighting, debugging, and extensions improve development efficiency. It also helps in organizing project files and testing the system smoothly.

4.3.9 Web Browser

A web browser like Google Chrome interacts and allows interface users publish their SMS messaging /call record classify them by check Scam or Safe in the browser. The basic features provided by web browsers include are JavaScript execution capabilities, multimedia capabilities and standard interactive features needs in order for the application run smoothly. Browsers also provide the connection between the user to the system; thus providing access and responses.

CHAPTER 5

SYSTEM IMPLEMENTATION

5.1 PROJECT DESCRIPTION

The SMS and Call Detection System that identifies Multilingual Scams is a machine learning, natural language processing solution designed to detect fraudulent messages and scam calls in real time by user security mechanisms that help a user to assess whether or not there is a potential threat before taking some type of action. The system provides a structured, data-driven method for categorizing SMS messages/calls into Three categories Safe, Suspicious or Scam and assigns a confidence score to each category which provides greater transparency and user trust.

The architecture of the system was developed in a modular fashion, and contains the following major components that are involved in detection and analysis: Data Input, Pre-processing, Feature Extraction, Classification, and Result Visualization. The data processing will be done in the background using the Python programming language; the methods used for processing the text, extracting features are TF-IDF; and the classification algorithm will be Naïve Bayes. All of the components will be interconnected so there is a seamless and efficient movement of the data through the detection process.

There are both call-based and SMS-based methods of detecting scams using this system. Scams can be detected through the submission of text messages to the system in the case of SMS-based detection, or by the system converting the audio into text format using Speech to Text technology in the case of call-based detection. The extracted text is then subject to further processing using various machine learning algorithms and techniques to identify whether or not the text is either a scam or legitimate. The dual-mode feature of the system supports more effective operations with real-world scam examples.

TF-IDF is used to extract significant features from text data, along with machine learning methods for identifying scam complexities. Additionally, this system supports multiple languages e.g., English, Tamil, Hindi, and includes an educational module used to provide consumers with an understanding of scams, scammers, prevention methods, and so forth, therefore enhancing digital literacy and security awareness for the entire world.

For storing data, the system utilizes SQLite to manage user input, processed messages, classification results and maintenance of system logs in an efficient manner. The application has been built to be lightweight, low-cost, and resource-efficient while providing accurate results. The overall goal of the project is to deliver a scalable, efficient, and user-friendly solution to real-time scam identification; therefore, users will be able to safeguard their private information and financial resources from continually changing cyber threats.

5.1.1 Methodology of the Project:

This project proposes a systematic method for detecting scam SMS messages and fraudulent call communications by using Natural Language Processing and machine learning algorithms. When users enter a message or upload a call recording through the web interface, it is sent to the backend system for processing. If the input is a call, speech-to-text technology converts it into text, which is then transformed into features using TF-IDF. These features are analyzed using a Naive Bayes model to classify the input as Safe, Suspicious, or Scam. The results are displayed to the user and stored in a database for future analysis.

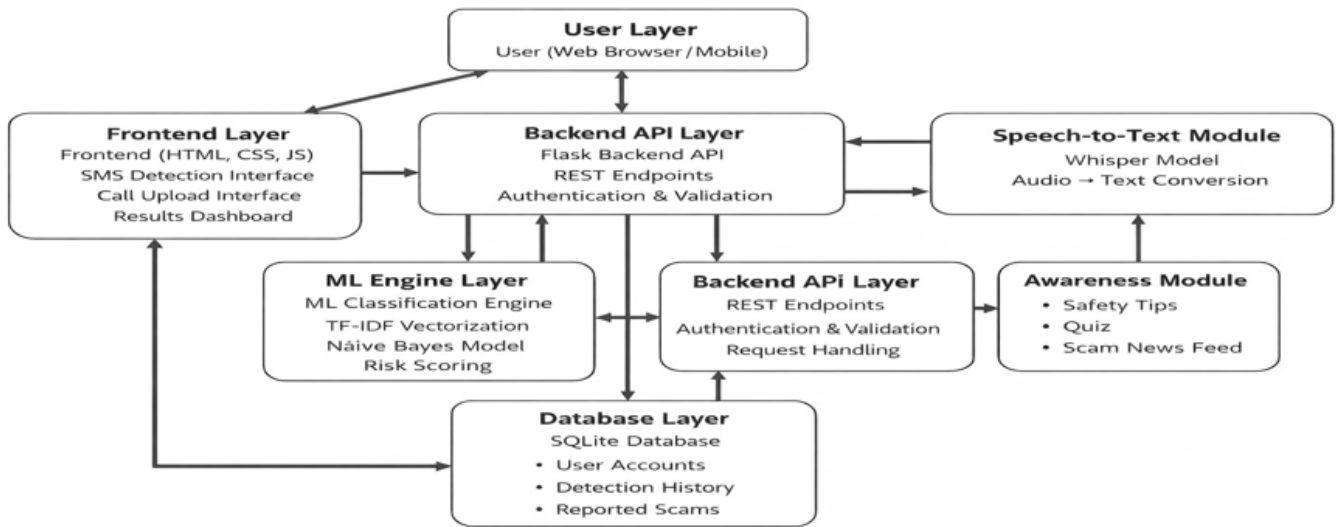


Figure 5.1 Methodology of the Project

5.2 MODULE DESCRIPTION

5.2.1 User Authentication Module

The User Interface Module provides the interface between users and the Multilingual Scam SMS and Call Detection System, allowing users to communicate easily with the system. Users can submit SMS messages or call transcripts to be analyzed for detection of potential scams and immediately receive results indicating the types of scams.

The User Interface Module displays the classification results e.g., Safe, Suspicious, or Scam with corresponding confidence levels; thus, the user is better able to determine if any action is needed. The User Interface Module additionally ensures that all user submissions are sent to the backend processing modules without errors and will provide the user with both clear alerts and results, which will help increase the user's awareness and safe usage of digital communication.

The User Interface Module is instrumental in enhancing the user's experience by displaying information in a clear, cohesive manner and allowing the user to quickly identify possible threats and take appropriate action. Overall, the User Interface Module serves as the primary method for users to interact with the Multilingual Scam SMS and Call Detection System and will enable users to effectively communicate with the system.

5.2.2 Message and Call Input Module

This module receives communication data text messages and voice calls from multiple data sources and provides an interface for users to enter the text messages directly and upload the audio file of their voice calls for analysis. The Module ensures that all input communication data is properly collected and ready for subsequent processing.

When voice-based input is received, the module leverages Speech-to-Text technology to convert the audio data to a text format for consistent analysis of spoken content with the same processing pipeline used to analyse text messages. The module provides proper formatting and validation of input communication data prior to sending it to the Preprocessing stage.

As this Module supports both text-based and voice-based input, it will improve the overall capability of the system to identify real-world scam scenarios. It will also provide flexibility and reliability with respect to data collection for the overall effectiveness of the system in identifying the different types of fraudulent communication.

5.2.3 Text Preprocessing Module

Cleaning and modifying raw text to create a standard structure for analysis is the Text Preprocessing Module's responsibility. It removes symbols, characters, and stop words that do not add meaning to the text or will not help in its interpretation. This setup tools eliminate noise from the raw input data and guarantees that only relevant pieces of information are sent for further work.

The Module will divide text into smaller pieces tokens to make analyzing easier. Tokenization makes this possible. The Module uses techniques for normalization to change the case of text to lowercase, eliminate punctuation, stem or lemmatize the text - to create standardization. These techniques will reduce variations among how a piece of text may be represented, increasing the level of consistency amongst text.

It converts unstructured text into a clean and structured format, which assists with increasing the machine learning model's performance. This process creates clean, organized input data from which to build features for model training, which will ultimately result in improved classification performance and overall system productivity.

5.2.4 Feature Extraction Module

The Feature Extraction Module takes raw preprocessed text and transforms it into numeric representations that can be used by machine learning algorithms. Using the Term Frequency-Inverse Document Frequency technique, it assigned weights to words based on the importance of each word within a single document, as well as the importance of each word across the entire data set. This allows us to identify the most meaningful keywords associated with suspicious or scam-like activities.

The TF-IDF technique will assign high relevance to words that are very pertinent to each individual message, while placing a lower degree of influence on frequently occurring words that do not aid in the classification of the messages. The feature vectors that are created from the TF-IDF process are capable of determining relevant patterns and relationships in the text data that are critical to making accurate predictions about the messages. The feature vectors will be passed to the classification module for additional analysis.

By converting text into relevant numeric data, the Feature Extraction Module provides the necessary information for effective analysis and interpretation of large amounts of data. The Feature Extraction Module plays a key role in enhancing the efficiency and accuracy of the overall detection process within the system architecture.

5.2.5 Scam Classification Module

The heart of the Fraud Detection System is the Scam Classification Module, which determines if incoming data is real or fraudulent. The module employs the Multinomial Naive Bayes classification approach a type of machine learning, which is effective for classifying textual data based on probability. The Multinomial Naive Bayes model is trained on labelled i.e., those identified as scams and not scams datasets to develop a fraud/non-fraud pattern.

The Scam Classification Module receives feature vectors created by the TF-IDF module, and uses these feature vectors to use probability to classify new data as Safe, Suspicious or Scam. The Scam Classification Module can detect multiple types of fraud and scams, such as phishing, fraud via financial messages, and malicious communication. Since the Naive Bayes approach to classification is lightweight, it can produce predictions extremely quickly, making it suitable for real-time applications.

The Scam Classification Module provides greater capabilities to accurately detect potential fraud and scam threats. It provides consistent performance while consuming minimal processing power, thus resulting in an efficient and scalable solution. The Scam Classification Module is a key component in accomplishing the main goal of this project - detecting scams in real time.

5.2.6 Database Management Module

The database management module is responsible for storing, organizing, and managing all data generated and processed by the entire system. The SQLite database system uses a lightweight, efficient, and relational database that doesn't require a separate server to operate. The database stores user input, SMS messages, call transcripts, classification results, and system logs in a structured way.

The database management module provides data integrity and data consistency by providing secure storage of information, allows efficient querying and retrieval of data to support seamless interaction among all components of the system, and supports the storage of historical records to be used for future analysis, assessment of performance, and system improvements.

The database management module provides reliable data storage and management that supports the overall infrastructure of the system's information. It provides real-time operations and ensures that all necessary data is available for analysis and reporting. In addition to the above, the database management module is fundamental to assuring the overall functionality and scalability of the system.

5.2.7 Administrative Monitoring Module

Administrators have the ability to manage and monitor how the operation runs on a daily basis. All actions taken by the operators are recorded and monitored by the administrator using this module.

Through this module, administrators will have the ability to review and analyze how well the system is accurately classifying scam messages or transcripts of phone calls. By monitoring and analyzing these metrics, an administrator will have the ability to evaluate how successfully the system has detected scam messages or transcripts of phone calls.

Additionally, the Administrative Monitoring Module provides administrators with the ability to identify new types of scam messages or calls that have been identified and to evaluate how successful the system has been at detecting and classifying those types of messages or calls. This module provides so much insight into how the system operates that an administrator can make informed decisions to change the detection strategy for better results. Another important feature of the Administrative Monitoring Module is the reporting and analysis capabilities. Administrators will be able to use these capabilities to generate very detailed reports to assist in evaluating results.

The Administrative Monitoring Module provides for ongoing, positive development and reliability of the system's operations. It allows for the effective management of the system's operations and provides for the ability of the system to evolve as new types of scams are identified. The Administrative Monitoring Module is a critical component to assist in maintaining the accuracy, quality, and reliability of the scam message and call detection process.

CHAPTER 6

RESULTS AND DISCUSSION

6.1 RESULTS

After being evaluated in a controlled environment to validate its performance, accuracy, and efficiency, the system successfully developed a Multilingual Scam SMS and Call Detection System which can perform analysis of SMS messages and telephone calls using natural language processing/ML technologies. The system expands the extent of detecting voice-based scams through incorporated Speech-to-Text capabilities beyond traditional text-only methods.

The use of TF-IDF as a feature extraction technique in conjunction with a Naive Bayes classification gorithm produced reliable and rapid classification results typed into one of three classes; Safe, Suspicious or Scam with a confidence score assigned for each. These confidence scores provide transparency to the users and increased trust in the system. Pre-processing techniques that were applied to SMS message sequences and telephone transcripts resulted in quality enhancing and noise reducing factors that improved overall prediction accuracy, along with achieving effective classification rates.

In addition to being utilized for evaluation purposes, the fraud awareness module provided instructional content to inform end users of fraud patterns and methods to avoid falling victim to those patterns. The system demonstrated low computational complexity and real-time detection capabilities making it suitable for use within the commercial marketplace. The results achieved through multi-country testing confirm that the system provides efficient, scalable, and effective detection of modern methods of scam communication.

6.2 SCREENSHOTS

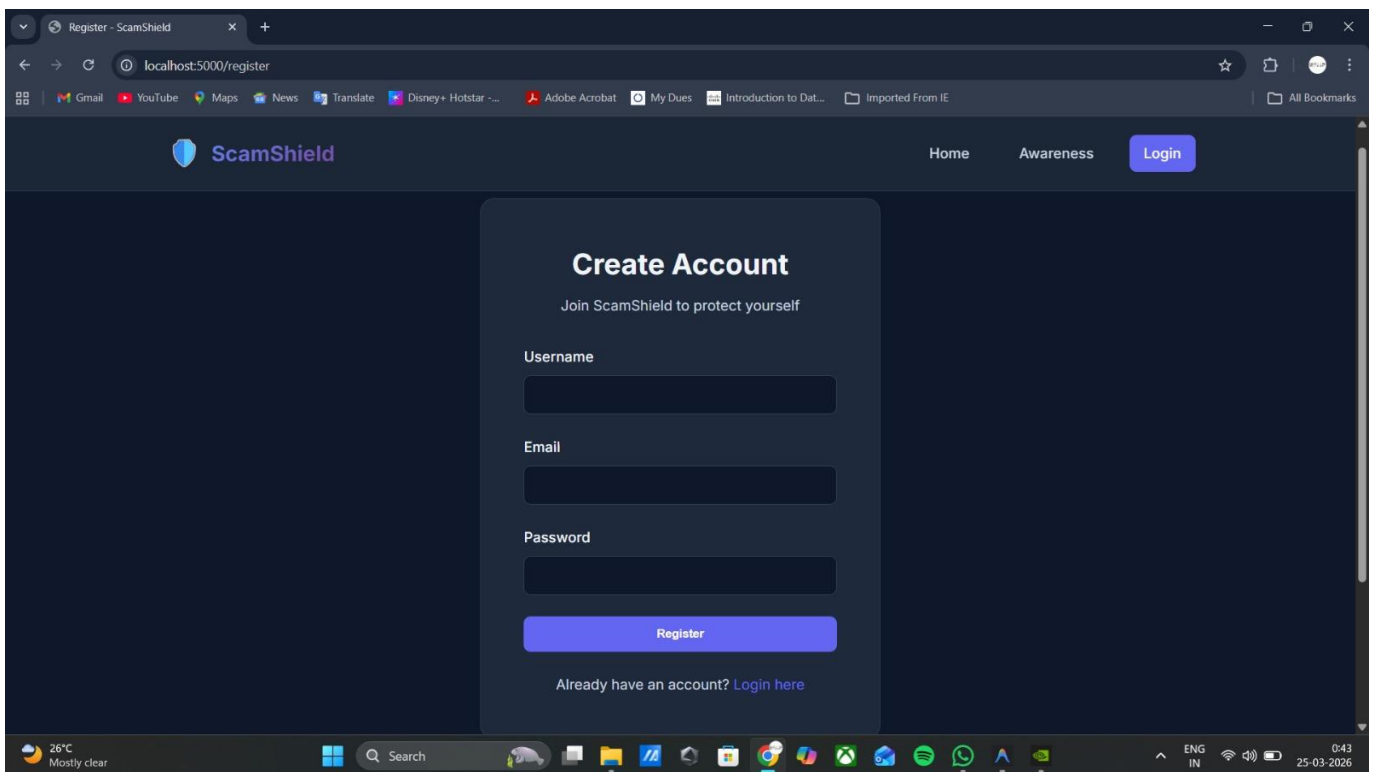


Figure 6.1 User Register

This Figure 6.1 describes the Registration page of my Multilingual Scam SMS and Call Detection System Scam Shield, where new users can create an account to access the system features. Users are required to enter their username, email, and password to register. This page provides a secure entry point for new users and ensures authentication before accessing scam detection services. It also includes a login option for existing users.

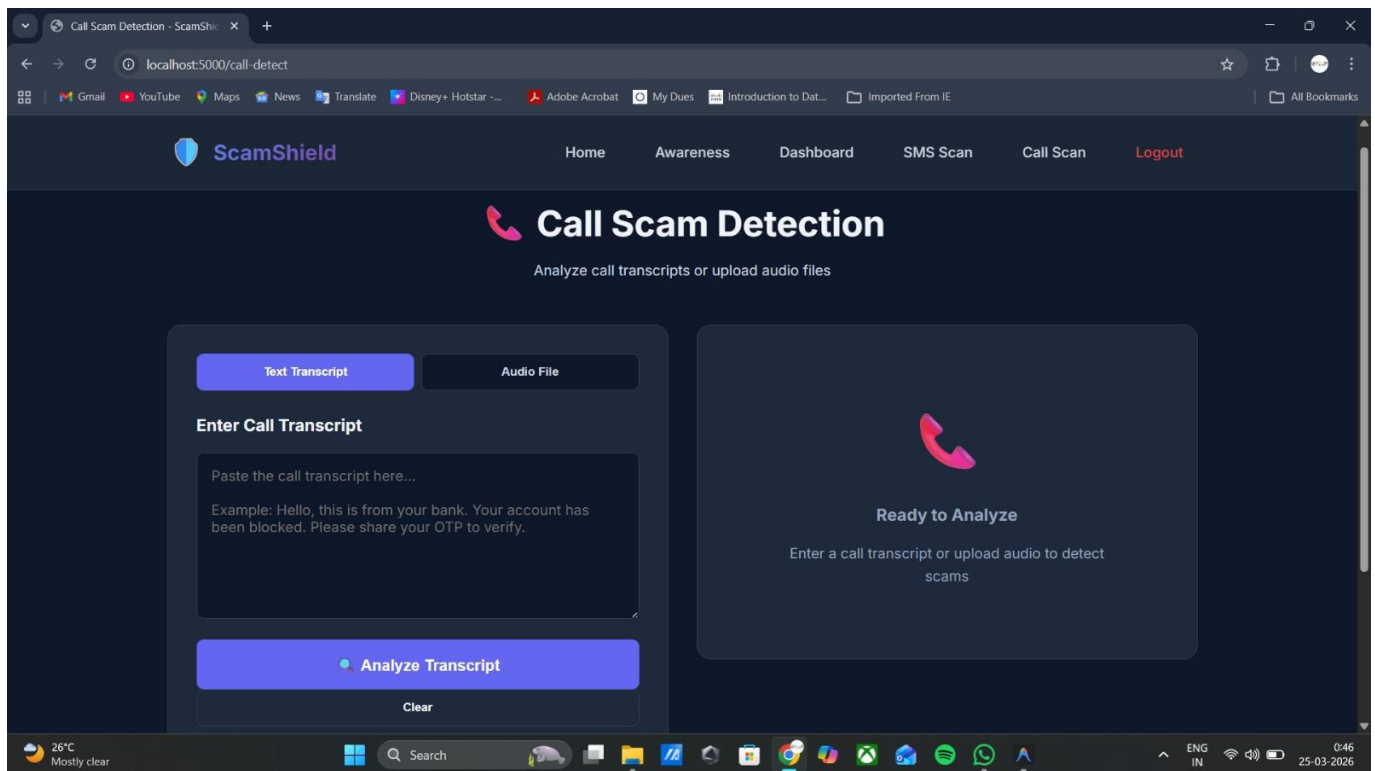


Figure 6.2 Call Scam Detection

This Figure 6.2 describes the Call Scam Detection page of my system, where users can analyze call transcripts or upload audio files to detect fraudulent calls. The system allows users to input call conversations, which are then processed using speech-to-text and machine learning techniques. Based on the analysis, the system classifies the call as Safe, Suspicious, or Scam, helping users identify voice-based fraud effectively.

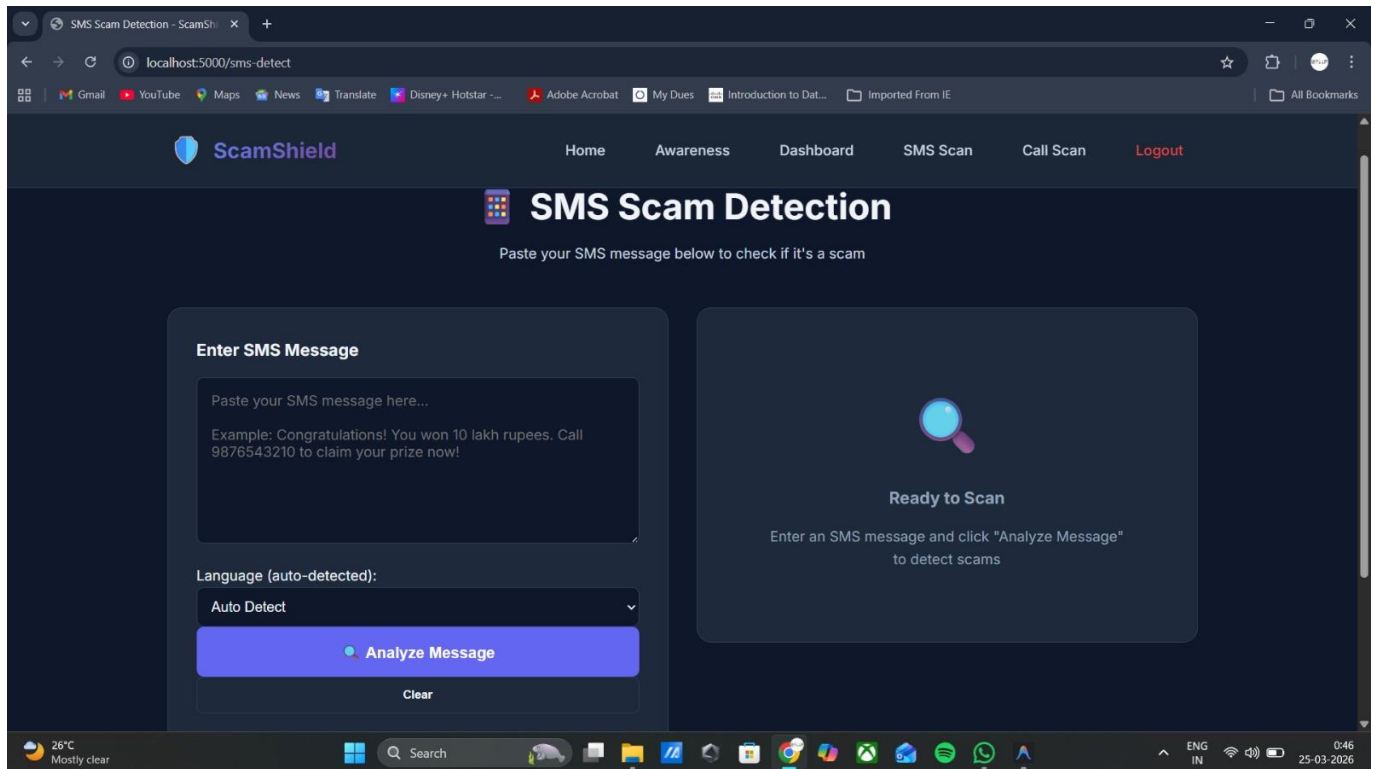


Figure 6.3 SMS Scam Detection

This Figure 6.3 describes the SMS Scam Detection page of my system, where users can enter SMS messages to check whether they are fraudulent or not. The system automatically detects the language and analyzes the message using NLP and machine learning models. It identifies scam patterns, suspicious links, and keywords, and provides a classification result to help users avoid SMS-based fraud.

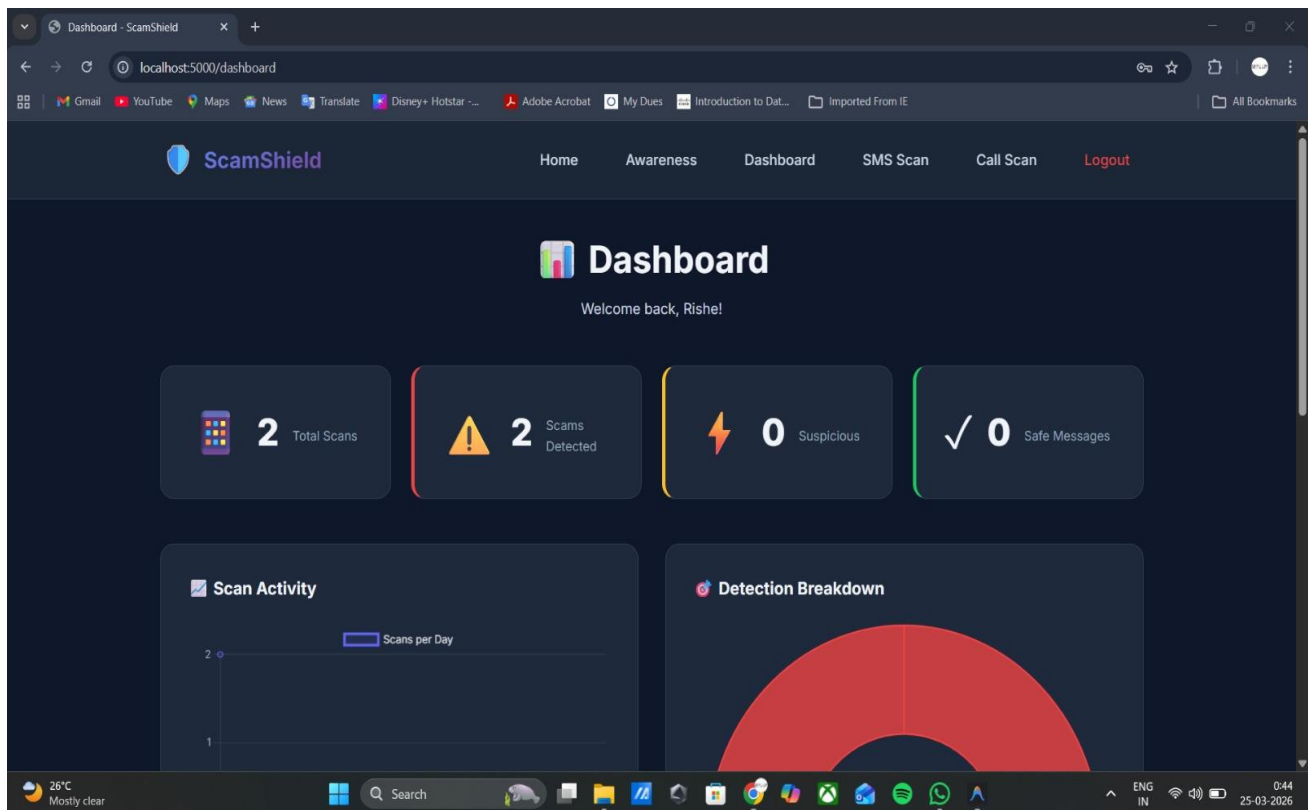


Figure 6.4 Dashboard

This Figure 6.4 describes the Dashboard page of my system, where users can view an overview of their scanning activity. It displays key statistics such as total scans, scams detected, suspicious messages, and safe messages. The dashboard also includes visual representations like graphs and charts to provide insights into detection results, helping users track and understand scam patterns efficiently.

CHAPTER 7

CONCLUSION AND FUTURE ENHANCEMENT

7.1 CONCLUSION

The Successfully developed and implemented an automated solution to assist with the increasing challenges related to sending fraudulent communications using machine learning and natural language processing, the Multilingual Scam SMS/Call Detection System applies TF-IDF method of extracting features from text SMS and data from both SMS messages & audio recordings calls that are processed using a Naive Bayes Classifier to categorize them as either Safe, Suspicious or Scam. This project has overcome restrictions of the former processing complementary solutions by using a real-time/synchronous process of detection at the user level instead of at the platform or system level of filtering.

The capability provided by integration of speech to text technology into the existing model extends the system's utility beyond the normal processing solely based upon text documents. Providing multilingual support English, Tamil, Hindi allows users of the system broader overall accessibility & use of the system. Additionally, providing confidence scores provides for transparency to the user about system predictions, and the results, validating trustworthiness of predictions. The awareness module provides users with more knowledge of what constituted scams and preventive measures against them.

The system demonstrated efficient processing performance with low computational requirements, making it a financially attractive and scalable solution. Overall, the system provides a logical solution assists end users in providing improved security for digital communications and improving user awareness.

7.2 FUTURE ENHANCEMENT

Future enhancements of the Multilingual Scam SMS and Call Detection System can focus on improving accuracy, scalability, and real-time performance by integrating advanced deep learning models such as transformer-based architectures for better contextual understanding of messages and calls. The system can be extended to support more regional and international languages, enabling wider usability across diverse user groups. Incorporating real-time alert mechanisms and mobile application integration can help users receive instant notifications for suspicious activities. Additionally, continuous learning techniques can be implemented to allow the model to adapt to evolving scam patterns over time. Enhancing the awareness module with interactive guidance and visual analytics can further educate users and improve decision-making. These improvements will make the system more robust, intelligent, and suitable for large-scale real-world deployment.

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APPENDIX-1

```
from flask import Flask, render_template, request, redirect, url_for, flash, jsonify,
session
from flask_sqlalchemy import SQLAlchemy
from flask_login import LoginManager, UserMixin, login_user, login_required,
logout_user, current_user
from werkzeug.security import generate_password_hash, check_password_hash
from datetime import datetime, timedelta
import os
import pickle
import re
import requests
import json
import csv
import io
from functools import wraps
import feedparser
from googletrans import Translator
import nltk
from nltk.corpus import stopwords
from nltk.tokenize import word_tokenize
from sklearn.feature_extraction.text import TfidfVectorizer
from sklearn.naive_bayes import MultinomialNB
import numpy as np

# Download required NLTK data
try:
    nltk.data.find('tokenizers/punkt')
```

```

except LookupError:
    try:
        nltk.download('punkt')
    except:
        nltk.download('punkt_tab')

try:
    nltk.data.find('corpora/stopwords')
except LookupError:
    nltk.download('stopwords')

app = Flask(__name__)
app.config['SECRET_KEY'] = 'your-secret-key-change-in-production'
app.config['SQLALCHEMY_DATABASE_URI'] = 'sqlite:///database.db'
app.config['SQLALCHEMY_TRACK_MODIFICATIONS'] = False

db = SQLAlchemy(app)
login_manager = LoginManager()
login_manager.init_app(app)
login_manager.login_view = 'login'

# Initialize translator
translator = Translator()

# Database Models
class User(UserMixin, db.Model):
    id = db.Column(db.Integer, primary_key=True)
    email = db.Column(db.String(120), unique=True, nullable=False)

```

```
password_hash = db.Column(db.String(128))
created_at = db.Column(db.DateTime, default=datetime.utcnow)

# Relationship
scans = db.relationship('Scan', backref='user', lazy=True)
quiz_scores = db.relationship('QuizScore', backref='user', lazy=True)
```

```
class Scan(db.Model):
    id = db.Column(db.Integer, primary_key=True)
    user_id = db.Column(db.Integer, db.ForeignKey('user.id'), nullable=False)
    message = db.Column(db.Text, nullable=False)
    result = db.Column(db.String(20), nullable=False) # safe, suspicious, scam
    score = db.Column(db.Float, nullable=False)
    language = db.Column(db.String(10))
    detected_keywords = db.Column(db.Text)
    scan_type = db.Column(db.String(20), default='sms') # sms, call
    created_at = db.Column(db.DateTime, default=datetime.utcnow)
```

```
class ReportedScam(db.Model):
    id = db.Column(db.Integer, primary_key=True)
    user_id = db.Column(db.Integer, db.ForeignKey('user.id'), nullable=False)
    scam_type = db.Column(db.String(50), nullable=False)
    description = db.Column(db.Text, nullable=False)
    phone_number = db.Column(db.String(20))
    created_at = db.Column(db.DateTime, default=datetime.utcnow)
```

```
class QuizScore(db.Model):
    id = db.Column(db.Integer, primary_key=True)
```

```

user_id = db.Column(db.Integer, db.ForeignKey('user.id'), nullable=False)
score = db.Column(db.Integer, nullable=False)
total_questions = db.Column(db.Integer, nullable=False)
percentage = db.Column(db.Float, nullable=False)
created_at = db.Column(db.DateTime, default=datetime.utcnow)

```

```
@login_manager.user_loader
```

```

def load_user(user_id):
    return User.query.get(int(user_id))

```

```
# ML Model Management
```

```
class ScamDetector:
```

```

    def __init__(self):
        self.model = None
        self.vectorizer = None
        self.scam_keywords = [
            'lottery', 'winner', 'prize', 'congratulations', 'claim', 'urgent',
            'bank', 'account', 'suspended', 'verify', 'otp', 'pin', 'password',
            'offer', 'discount', 'free', 'gift', 'reward', 'bonus', 'cashback',
            'investment', 'profit', 'guaranteed', 'risk-free', 'limited time',
            'click here', 'link', 'website', 'download', 'install', 'update',
            'payment', 'transaction', 'refund', 'billing', 'invoice', 'due',
            'immediate', 'action required', 'expires', 'today', 'now'
        ]

```

```
def load_model(self):
```

```
    """Load pre-trained model and vectorizer"""
```

```
    try:
```

```

with open('models/sms_model.pkl', 'rb') as f:
    self.model = pickle.load(f)
with open('models/vectorizer.pkl', 'rb') as f:
    self.vectorizer = pickle.load(f)
return True
except FileNotFoundError:
    return False

def train_model(self):
    """Train a new model with sample data"""
    # Sample training data
    scam_samples = [
        "Congratulations! You have won ₹10 lakh in our lottery. Click here to claim your prize.",
        "Your bank account has been suspended. Verify your details immediately to reactivate.",
        "URGENT: Your OTP is 123456. Do not share with anyone.",
        "Free iPhone 15! Click the link to claim your gift now.",
        "Investment opportunity: Guaranteed 50% returns in 30 days. Limited time offer.",
        "Your payment of ₹5000 is due today. Pay now to avoid late fees.",
        "You have been selected for a special offer. Act now before it expires.",
        "Download our app to get ₹1000 cashback instantly.",
        "Your account will be closed if you don't verify within 24 hours.",
        "Congratulations! You are the lucky winner of our mega contest."
    ]

```

```

safe_samples = [
    "Your electricity bill of ₹1200 is due tomorrow. Please pay on time.",
    "Your package has been delivered successfully. Thank you for shopping with
us.",
    "Your appointment with Dr. Smith is scheduled for tomorrow at 2 PM.",
    "Weather update: It will be sunny today with temperature around 28°C.",
    "Your subscription to Netflix will expire in 3 days. Renew to continue
watching.",
    "Thank you for your recent purchase. Your order will be shipped within 2
days.",
    "Your gym membership expires next month. Contact us to renew.",
    "Your flight to Mumbai is scheduled for departure at 6:30 AM.",
    "Your insurance premium of ₹5000 is due next week.",
    "Your internet bill of ₹800 has been paid successfully."
]

```

```

# Prepare training data

```

```

texts = scam_samples + safe_samples
labels = ['scam'] * len(scam_samples) + ['safe'] * len(safe_samples)

```

```

# Create vectorizer and transform texts

```

```

self.vectorizer = TfidfVectorizer(max_features=1000, stop_words='english')
X = self.vectorizer.fit_transform(texts)

```

```

# Train model

```

```

self.model = MultinomialNB()
self.model.fit(X, labels)

```

```

# Save model and vectorizer

```

```

os.makedirs('models', exist_ok=True)
with open('models/sms_model.pkl', 'wb') as f:
    pickle.dump(self.model, f)
with open('models/vectorizer.pkl', 'wb') as f:
    pickle.dump(self.vectorizer, f)

return True

def preprocess_text(self, text):
    """Preprocess text for analysis"""
    # Convert to lowercase
    text = text.lower()

    # Remove special characters and numbers
    text = re.sub(r'^a-zA-Z\s', '', text)

    # Tokenize and remove stopwords
    tokens = word_tokenize(text)
    stop_words = set(stopwords.words('english'))
    tokens = [token for token in tokens if token not in stop_words]

    return ' '.join(tokens)

def detect_keywords(self, text):
    """Detect scam-related keywords in text"""
    text_lower = text.lower()
    detected = []
    for keyword in self.scam_keywords:

```

```

        if keyword in text_lower:
            detected.append(keyword)
    return detected

def predict(self, text):
    """Predict if text is scam or safe"""
    if not self.model or not self.vectorizer:
        if not self.load_model():
            self.train_model()

    # Preprocess text
    processed_text = self.preprocess_text(text)

    # Transform text
    X = self.vectorizer.transform([processed_text])

    # Predict
    prediction = self.model.predict(X)[0]
    probability = self.model.predict_proba(X)[0]

    # Get scam probability
    scam_prob = probability[0] if prediction == 'scam' else probability[1]

    # Detect keywords
    keywords = self.detect_keywords(text)

    # Determine risk level
    if scam_prob >= 0.7:

```

```

        result = 'scam'
    elif scam_prob >= 0.3:
        result = 'suspicious'
    else:
        result = 'safe'

    return {
        'result': result,
        'score': scam_prob * 100,
        'keywords': keywords,
        'explanation': f"Detected keywords: {' ', '.join(keywords) if keywords else
'None'}"
    }

```

Initialize detector

```
detector = ScamDetector()
```

Routes

```
@app.route('/')
def index():
```

```
    return render_template('index.html')
```

```
@app.route('/register', methods=['GET', 'POST'])
```

```
def register():
```

```
    if request.method == 'POST':
```

```
        email = request.form['email']
```

```
        password = request.form['password']
```

```

    if User.query.filter_by(email=email).first():
        flash('Email already registered!')
        return redirect(url_for('register'))

    user = User(email=email,
password_hash=generate_password_hash(password))
    db.session.add(user)
    db.session.commit()

    flash('Registration successful! Please login.')
    return redirect(url_for('login'))

return render_template('register.html')

@app.route('/login', methods=['GET', 'POST'])
def login():
    if request.method == 'POST':
        email = request.form['email']
        password = request.form['password']

        user = User.query.filter_by(email=email).first()
        if user and check_password_hash(user.password_hash, password):
            login_user(user)
            return redirect(url_for('dashboard'))
        else:
            flash('Invalid email or password!')

return render_template('login.html')

```

```
@app.route('/logout')
```

```
if __name__ == '__main__':
```

```
    with app.app_context():
```

```
        db.create_all()
```

```
        # Create demo user if not exists
```

```
        if not User.query.filter_by(email='demo@example.com').first():
```

```
            demo_user = User(
```

```
                email='demo@example.com',
```

```
                password_hash=generate_password_hash('demo123')
```

```
            )
```

```
            db.session.add(demo_user)
```

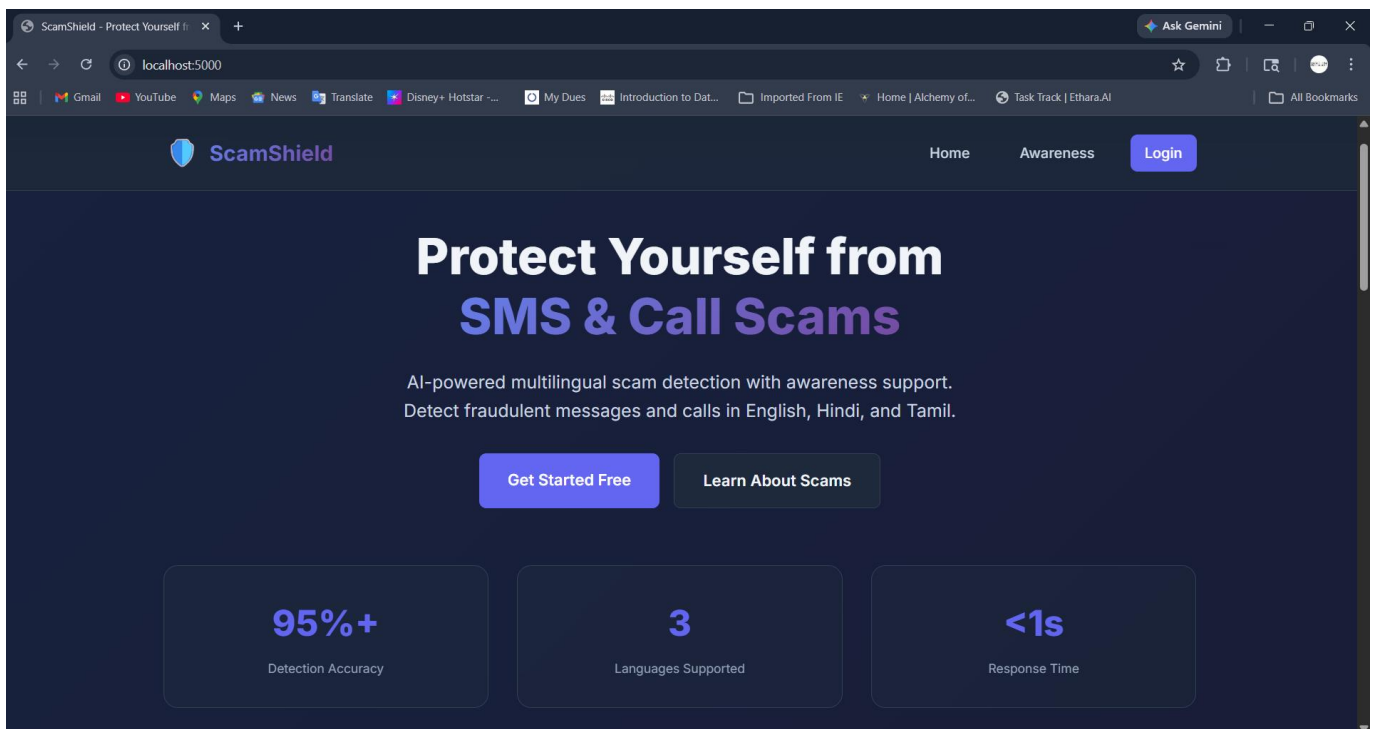
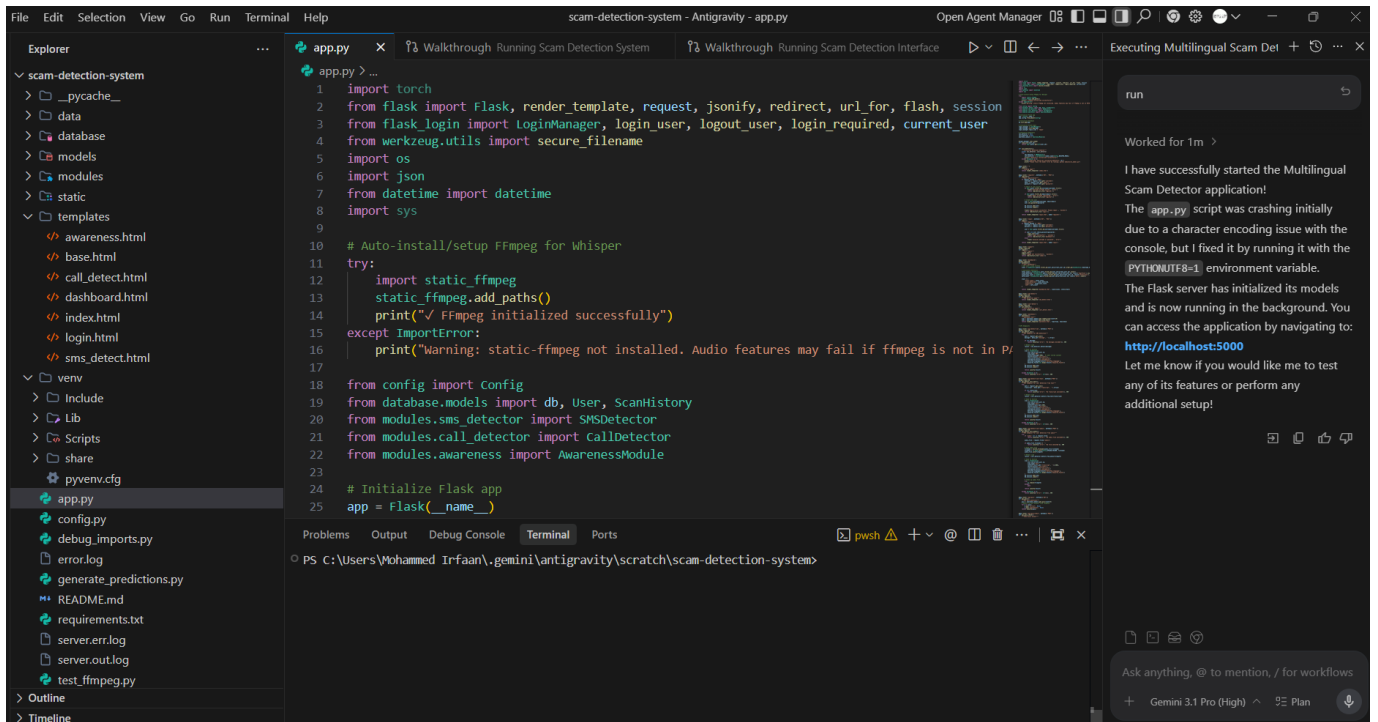
```
            db.session.commit()
```

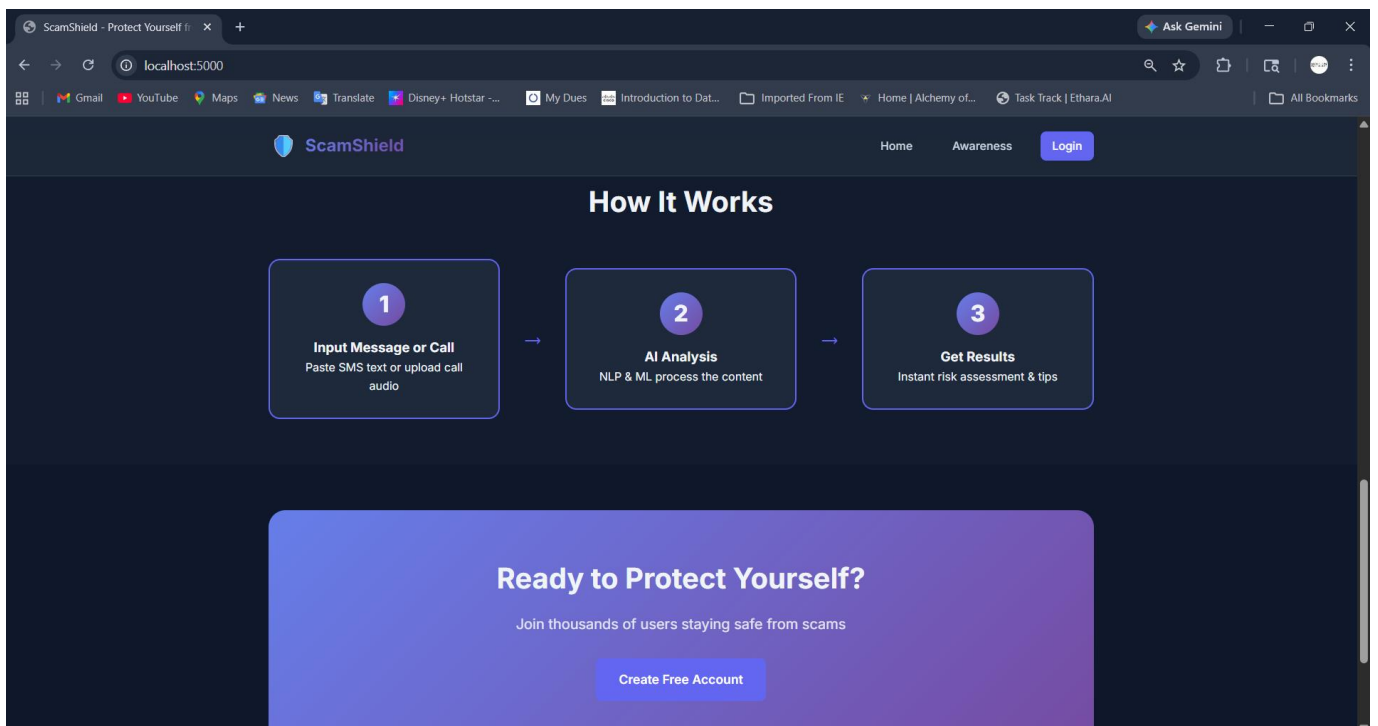
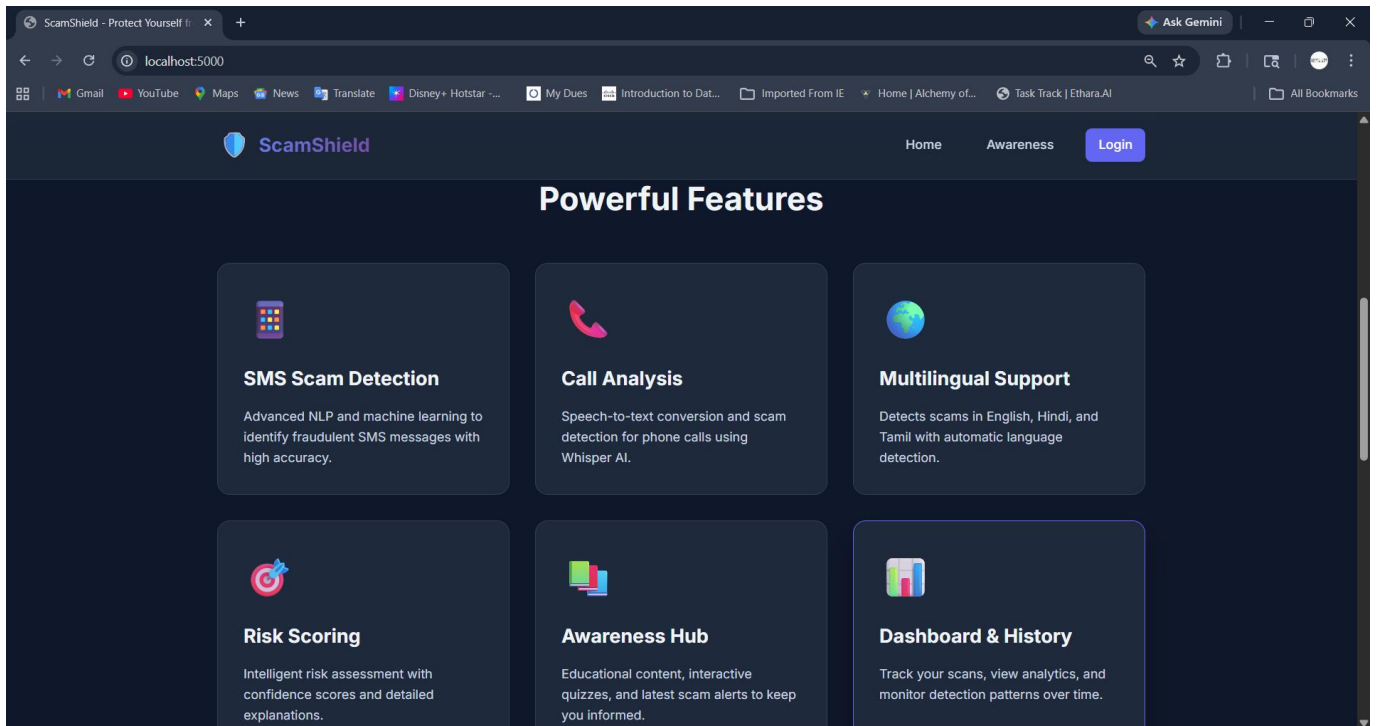
```
app.run(debug=True, host='0.0.0.0', port=5000)
```

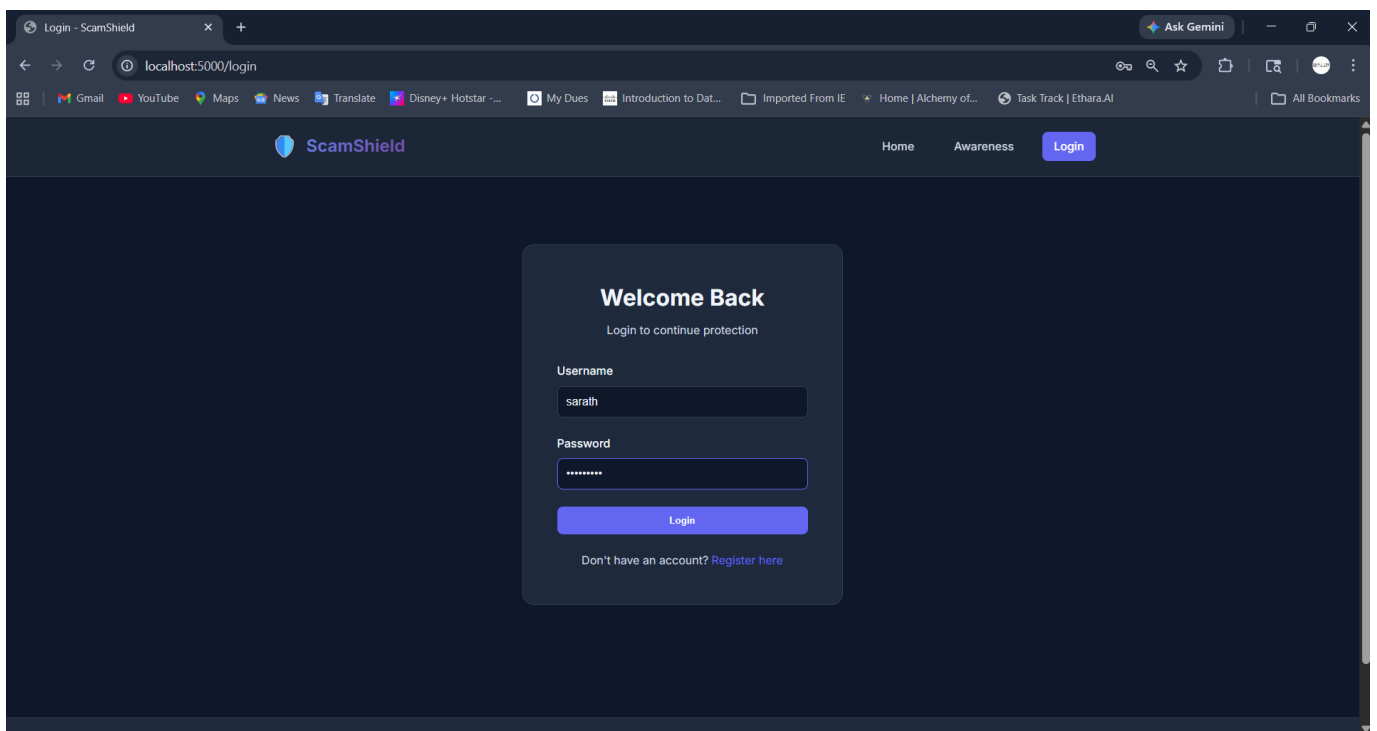
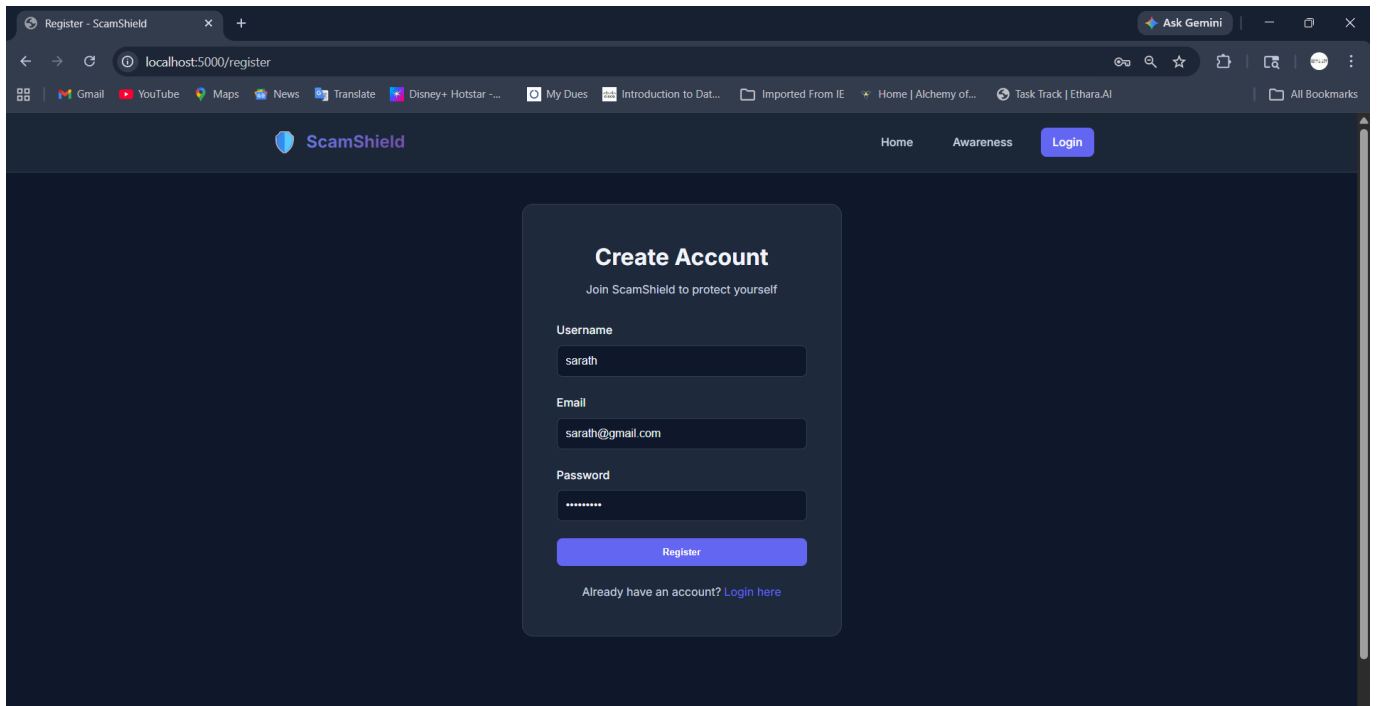
APPENDIX-2

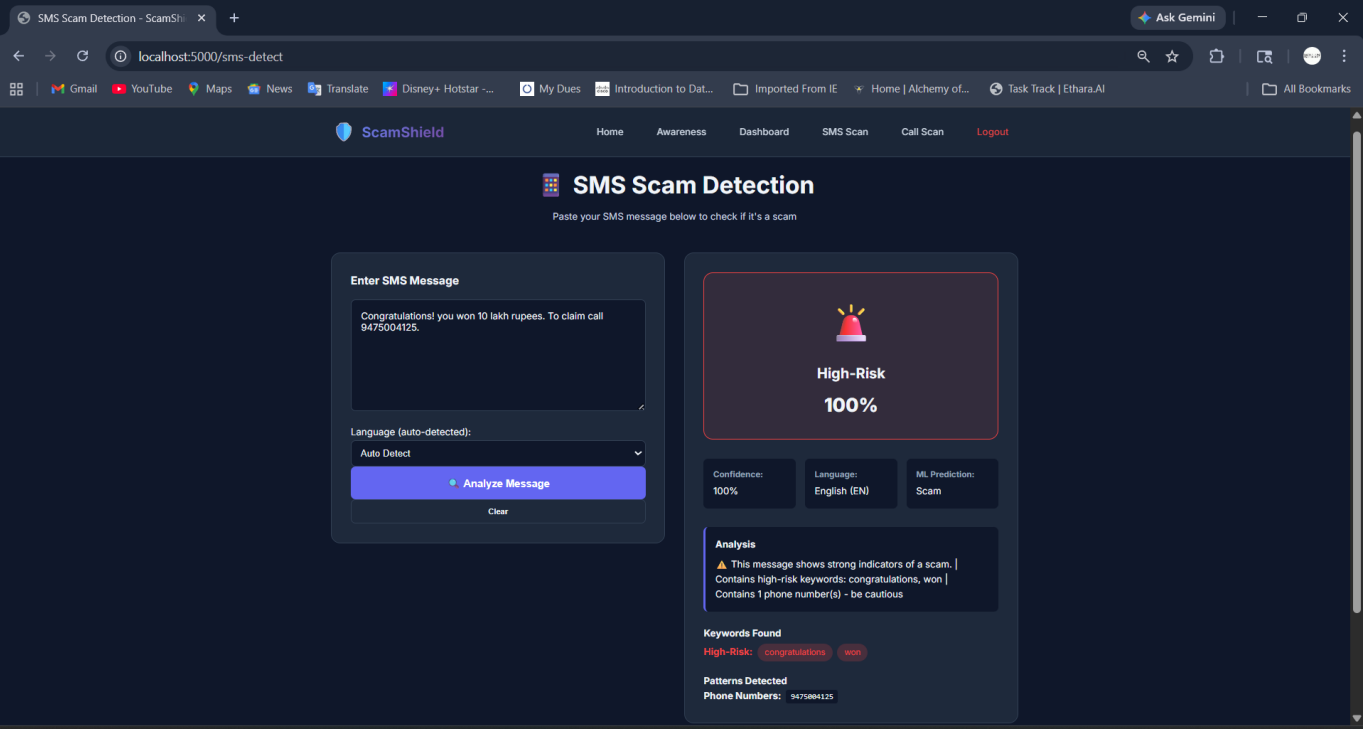
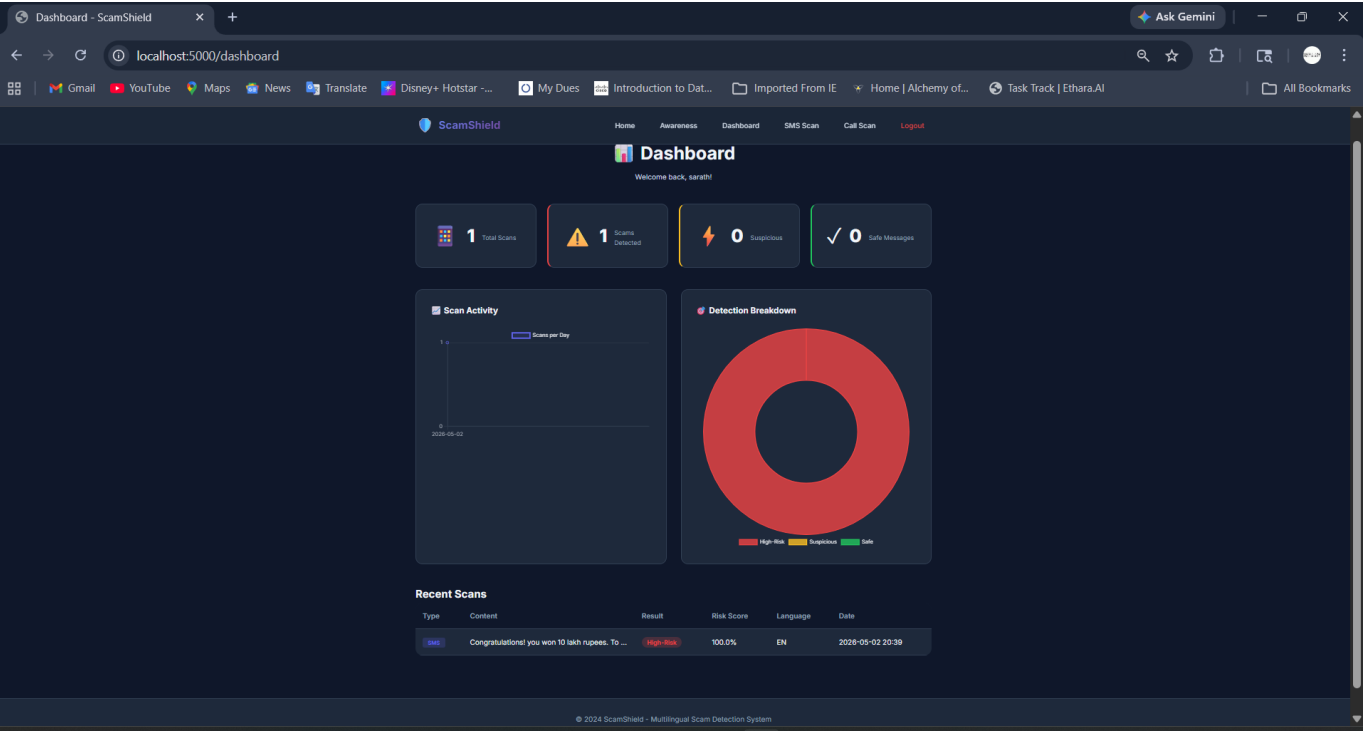
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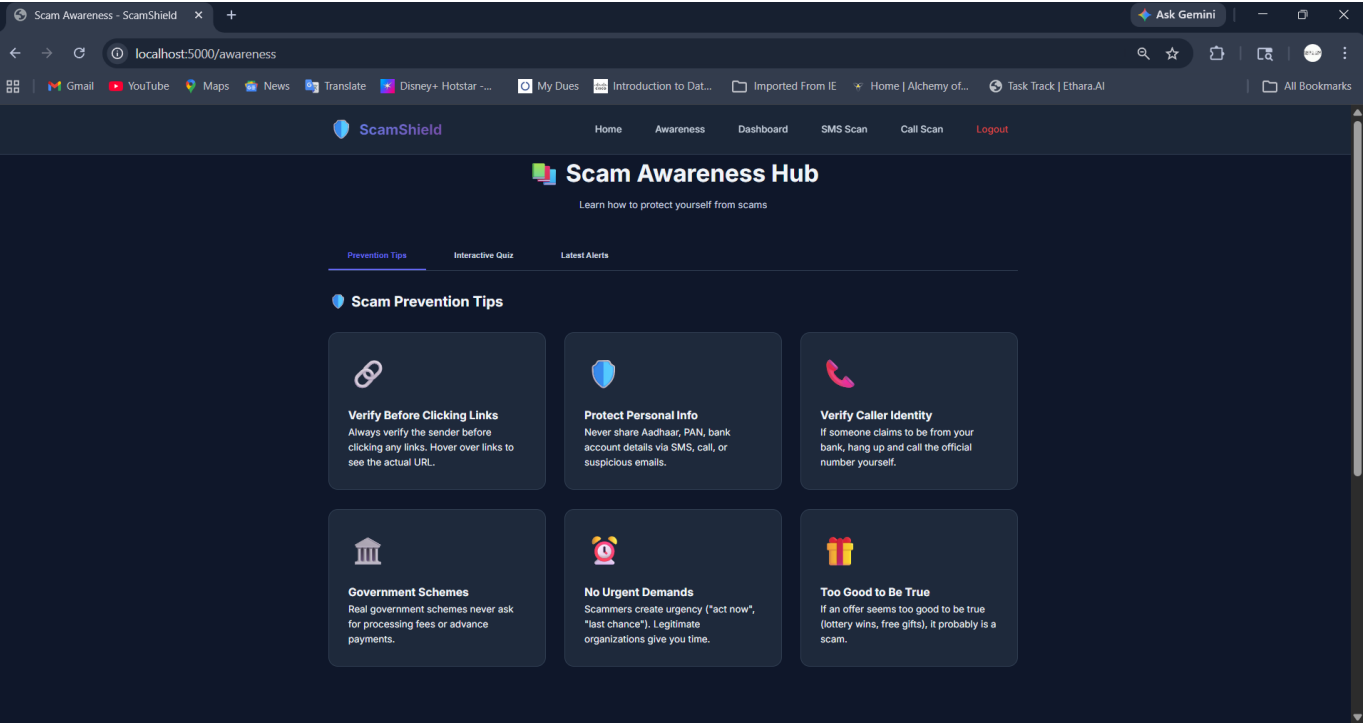
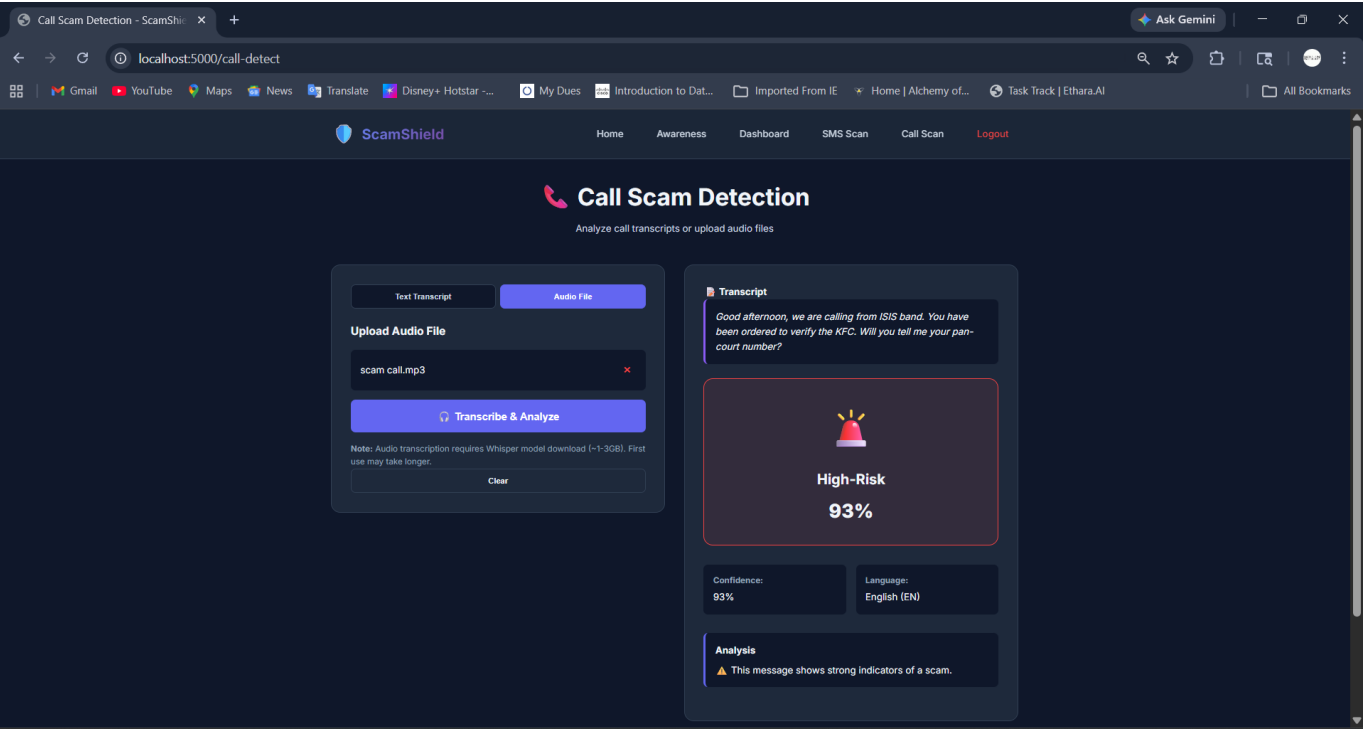
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Dear Author,

Thank you for your interest in the 7th International Conference on Artificial Intelligence and Smart Environments (ICAISE'2025). We are pleased to inform you that your paper has been accepted for oral presentation at ICAISE'2025. Your paper will be published in the conference proceedings by Springer under the series Lecture Notes in Networks & Systems (indexed in SCOPUS, etc.).

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for having attended the 7th International Conference on Artificial Intelligence and Smart Environments "ICAISE'2025", held from November 6 to 8, 2025, in Hammamet, Tunisia and presented his communication entitled:

" Customer Churn Prediction: From Data to Decision"

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for having attended the 7th International Conference on Artificial Intelligence and Smart Environments "ICAISE'2025", held from November 6 to 8, 2025, in Hammamet, Tunisia and presented his communication entitled:
" Smart Travel Companion System with Budget Op-timization Using Recommendations "

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